

Fingerprinting Encrypted DNS: Exploiting Metadata Leakage in DNS over QUIC

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Abstract—The growing adoption of HTTP/3 and its underlying transport protocol, Quick UDP Internet Connections (QUIC), represents a major step forward in Internet communications, improving performance, latency, congestion control, and encryption. Domain Name Server (DNS) over QUIC (DoQ) has emerged to enhance DNS confidentiality and performance, but remains vulnerable to website fingerprinting (WF) attacks, which exploit observable traffic patterns, even in the presence of encryption. In this study, we conduct a detailed analysis into the vulnerability of the DoQ protocol to WF attacks, examining how encrypted DNS traffic can still leak identifiable patterns that adversaries may exploit to infer users' web activity. We implement a comprehensive feature extraction framework adapted to encrypted DNS traffic. This approach enables accurate classification using a real-world dataset. We apply explainable Artificial Intelligence (XAI) to identify which features drive model predictions, offering new insights into the sources of metadata leakage. The findings of the present study reinforce the efficacy of fingerprinting attacks on DoQ traffic, highlighting the persistent nature of these vulnerabilities despite DoQ's encryption mechanisms. Consequently, the development of effective fingerprinting mitigation strategies in encrypted environments like DoQ continues to pose a critical challenge for protecting user privacy, underscoring the need for robust mitigation strategies to safeguard user privacy.

Index Terms—Encrypted DNS; Traffic Analysis; User Privacy; Website Fingerprinting

I. INTRODUCTION

The growing adoption of HTTP/3 and its transport protocol, Quick UDP Internet Connections (QUIC), represents a major step forward in modern Internet communications, offering notable improvements in performance, latency, congestion control, and integrated encryption. These improvements have enabled new encrypted Domain Name Server (DNS) protocols, such as DNS over QUIC (DoQ), which leverage QUIC's efficiency to improve DNS privacy and responsiveness. As a result, DoQ is gaining traction in mobile and ubiquitous networks, where latency and performance are critical.

However, despite DoQ's notable advantages over traditional unencrypted DNS and even over DNS over TLS (DoT) and DNS over HTTPS (DoH) in certain respects, it is not exempt from emerging privacy risks. In particular, DoQ remains vulnerable to website fingerprinting (WF) attacks, which exploit observable traffic patterns such as packet sizes, inter-arrival times, and stream behaviors, to infer visited domains, even

when content is encrypted. Existing literature has demonstrated that WF attacks can achieve high classification accuracy and can reveal sensitive user information [1]. Moreover, the centralization of DNS resolution in public resolvers and cloud infrastructures amplifies the risks of traffic correlation, surveillance, and dependency, especially in sensor networks and ad hoc wireless deployments [2].

Several mitigation strategies have been proposed, including DNS payload padding and artificial response delays, yet these countermeasures often come at the cost of increased latency or degraded user experience. Furthermore, while improvements in stream scheduling and transport-layer multiplexing have enhanced fairness and responsiveness in QUIC [3], recent studies indicate that DoQ still poses unresolved trade-offs between performance and privacy [4].

In this study, we conduct a detailed analysis into the vulnerability of the DoQ protocol to WF attacks, examining how encrypted DNS traffic can still leak identifiable patterns that adversaries may exploit to infer user activity and compromise privacy. In summary, the main contributions are: (i) we provide a targeted analysis of WF threats specific to DoQ, extending beyond prior work on DoT, DoH, or generic QUIC traffic; (ii) we develop a feature extraction framework tailored to encrypted DNS flows, enabling high-accuracy classification using a dataset; (iii) we demonstrate that WF remains effective under realistic open-world conditions, highlighting persistent metadata leakage despite encryption; (iv) we leverage explainable Artificial Intelligence (XAI) to uncover which traffic features most influence classification decisions; and (v) we offer actionable insights for the design of future DoQ countermeasures by quantifying privacy-performance trade-offs.

The remainder of the paper is structured as follows: Section II presents an overview of the encrypted DNS protocol while Section III reviews important aspects of encrypted network traffic classification focusing on the WF in related work. An overview of the encrypted website fingerprinting methodology is presented in Section IV. Section V discusses the traffic analysis conducted. Then, Section VI summarizes the results and the discussion derived from this study. Finally, Section VII draws the conclusions obtained.

II. ENCRYPTED DNS

The DNS has evolved significantly from its original plaintext form to more secure, encrypted protocols. DNS uses plaintext over port 53, exposing user queries to third parties. Traditional DNS is vulnerable to interception and can expose user queries, posing privacy and security risks. To address these concerns, DoT and DoH were introduced in 2016 and 2018, respectively. Both protocols improve user privacy by encrypting DNS queries. However, they also introduce new challenges. In particular, DoT and DoH rely on Transmission Control Protocol (TCP), which adds latency and remains susceptible to WF attacks [5].

As a natural progression, DoQ was proposed and standardized by the IETF in 2022 (RFC 9250 [6]). DoQ leverages QUIC as the underlying transport protocol. QUIC is a modern protocol built on top of the User Datagram Protocol (UDP) that integrates Transport Layer Security (TLS) 1.3, supports multiplexed streams, enables 0-RTT connection establishment, and provides better resistance to packet loss. Compared to DoT and DoH, the main advantage of DoQ is its lower latency, made possible by the 0-RTT capability (see Fig. 1). This renders DoQ comparable in performance to traditional DNS over UDP, while still offering encryption and enhanced reliability.

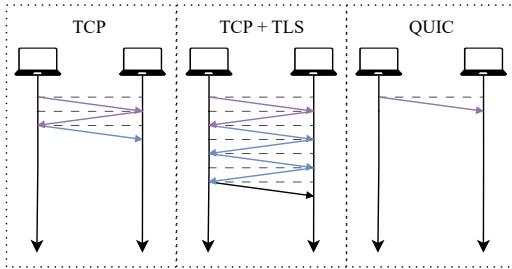


Fig. 1. Zero RTT Connection Establishment of QUIC protocol.

When a client initiates a DoQ connection, it begins with a QUIC handshake that includes the negotiation of TLS 1.3, ensuring end-to-end encryption from the start. Once established, a secure DoQ session is established, enabling reliable transmission of DNS queries and responses. Each query-response pair is transmitted over a separate unidirectional QUIC stream within the same session, allowing concurrent transmission of multiple DNS messages.

DoQ encrypts not only the DNS payload but also transport metadata such as packet headers (as illustrated in Fig. 2), providing enhanced protection against passive monitoring and fingerprinting relative to earlier encrypted DNS protocols. While the default port for DoQ is 853, servers can operate on alternative ports by prior agreement between client and server. Although the content and headers are encrypted, certain metadata, such as packet sizes, timing information, and stream behavior, remains exposed.

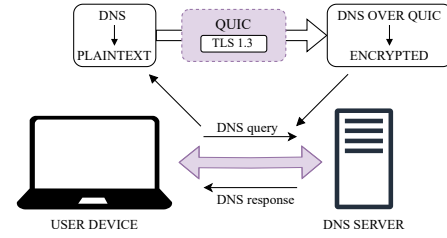


Fig. 2. DoQ Workflow.

III. RELATED WORK

The architectural advancements of QUIC introduce novel security and privacy challenges. While prior research has predominantly focused on QUIC's performance benefits, its security and privacy implications remain comparatively underexplored. Recent studies [7] offer a systematic exploration of QUIC's threat landscape, addressing cryptographic foundations, attack surfaces, and emerging privacy challenges. This section reviews existing works related to WF attacks on encrypted traffic, with a focus on QUIC and encrypted DNS, specifically DoQ, as well as prior efforts to mitigate such vulnerabilities.

A. Website Fingerprinting Attacks

WF is a type of traffic analysis attack that seeks to infer user activity, such as visited websites, by analyzing features of network traffic. Even if the traffic is encrypted, these attacks can still be carried out because there is still metadata that can be analyzed. Liu et al. [8] provide a comprehensive survey of WF techniques across various protocols.

Smith et al. [9] introduced a QUIC-TCP WF dataset and demonstrated that classifiers trained solely on TCP traffic performed poorly on QUIC traces, missing up to 96% of visits to monitored sites. Luxemburk et al. [10] evaluated machine learning classifiers, including Convolutional Neural Networks (CNNs) and LightGBM, using the CESNET-QUIC22 dataset, achieving up to 88% accuracy in service-level identification in QUIC traffic. Ha and Roch [11] proposed a framework for fingerprinting encrypted QUIC traffic. The system automated both feature selection and model training, achieving over 90% accuracy in closed-world scenarios using classifiers such as gradient boosting, Support Vector Machines (SVMs), and CNNs.

Siby et al. [12] conducted a comprehensive evaluation of padding-based defenses against WF in the context of QUIC traffic, focusing on scenarios where users browse the web directly without anonymity overlays such as VPNs or Tor. The authors found that even in the presence of encryption and padding, WF attacks remain effective unless additional countermeasures are implemented.

B. Encrypted DNS

Recent studies have extended their focus to encrypted DNS protocols such as DoH, which aim to enhance privacy but still

exhibit traffic patterns [13]–[15]. Within this context, DoQ has emerged as a subject of growing interest, with several studies examining its performance characteristics alongside its security and privacy implications. Notably, Arana et al. [16] discussed implementation considerations of DoQ and its interoperability with HTTP/3. Recent studies [17], [18] have evaluated DoQ integration with HTTP/3, showing improved DNS resolution latency. Specifically, Sengupta et al. [18] reported web page load time reductions of 33% on fixed networks and 50% on mobile networks by leveraging 0-RTT connection coalescing.

From a privacy perspective, Csikor et al. [1] released a benchmark dataset of web traffic over QUIC and developed transformer-based models to attack both DoQ and HTTP/3 traffic. Their findings suggest that DoQ, while encrypted, still leaks metadata exploitable for WF.

Hu and Fukuda [19] analyzed privacy leakage from DoQ by testing traffic from two recursive resolvers: NextDNS and Bind. They demonstrated that adversaries can infer whether websites fall into sensitive categories—such as health or finance—by using features like inter-packet arrival time and packet sizes. The authors tested two countermeasures: DNS payload padding and randomized response delays. Delays reduced F1 scores by 21% and 19% on NextDNS and Bind, respectively, while padding alone achieved a 10% reduction.

In conclusion, existing research demonstrates that although QUIC and DoQ provide substantial enhancements in performance and security, they exhibit distinct traffic patterns that render them susceptible to WF attacks. These findings highlight the necessity for analytical approaches to accurately assess the privacy implications of DoQ and emphasize the ongoing need for effective, protocol-specific mitigation strategies to address its inherent vulnerabilities.

IV. ENCRYPTED FINGERPRINTING

This section presents the foundational components of our framework for evaluating WF on DoQ traffic.

A. Threat model

We assume a threat model with defined adversarial capabilities, assets of interest, goals, assumptions, and the threat surface relevant to WF in the context of DoQ traffic.

1) *Adversary Capabilities*: We assume a passive network-level adversary who has access to observe client-side traffic without injecting, modifying, or dropping packets. The attacker (i) has prior knowledge with access to labeled traffic traces for a fixed set of monitored websites (closed-world setting), and (ii) in the open-world setting, the adversary must distinguish these from visits to unknown sites using limited or no training data for the unmonitored class.

2) *Assets/Targets*: The primary target of the adversary is the user’s browsing behavior, particularly the identity of visited websites, despite encryption of the content and domain names.

3) *Adversary Goals*: The adversary’s goals include: (i) accurately identifying which monitored site the user is visiting, assuming the true site is within the known set (closed-world classification); (ii) determining whether the visited site is monitored or unknown (open-world detection); and (iii) inferring user activity even when payloads and domain names are encrypted.

4) *Assumptions*: The threat model operates under the following assumptions: (i) packet payloads (QUIC, DoQ, HTTP/3) are fully encrypted; and (ii) the client does not employ advanced traffic obfuscation mechanisms.

5) *Threat Surface*: Several features of DoQ expose metadata that can be exploited by a fingerprinting adversary: size-based and timing-based leakage patterns may differ per site.

6) *Potential Impacts*: WF attacks in the context of encrypted protocols can impact user privacy and network security, despite the encryption at the transport and application layers.

a) *Privacy Impacts*: By exploiting features, such as packet size, timing, and flow patterns, WF attacks can identify visited websites, compromising confidentiality in both closed- and open-world settings and enabling user de-anonymization when combined with metadata.

b) *Security Impacts*: WF enables censorship and surveillance by selective blocking encrypted traffic, which undermines trust in protocols by leaking behavioral metadata, which can enable traffic correlation attacks between sessions or users.

B. Experimental Setup

- **Dataset**: We select the top 100 websites ordered by Tranco¹ rank, a stable website popularity list aggregated from multiple sources for research purposes.
- **Feature extraction**: We collect statistical metrics from packet-level traffic traces.
- **Models used**: We evaluate multiple supervised machine learning models, such as Random Forest, KNN, and LightGBM, to classify website visits based on traffic features.
- **Train-test split**: Initially, the dataset is partitioned into training and testing subsets using an 80/20 stratified split according to the domain label, ensuring proportional representation of each class, followed by class balancing.
- **Performance metrics**: Model performance is assessed using accuracy, precision, recall, and F1-score to capture both overall classification quality and class-specific effectiveness.

C. Dataset Details

Dataset of network traffic is used in this study, specifically DoQ+QUIC web traffic dataset captured by Csikor et al. [1]. It captures real-world network traffic collected during visits to QUIC-enabled websites with DoQ resolution. Traffic was collected from diverse locations using the CloudLab testbed, ensuring geographic and network diversity.

¹<https://tranco-list.eu/>

The dataset consists of directories labeled after visited domains, each containing CSV files of packet-level traces representing individual sessions. Network traffic includes fields such as raw header metadata (source/destination IPs and ports), protocol type (DoQ or not), packet size, inter-arrival time of two consecutive packets, and packet direction (incoming/outgoing). Table I presents the details regarding the number of websites and the number of sessions per each website for the different location points (Clemson, Massachusetts, Utah, and Wisconsin).

TABLE I
DATASET DETAILS FOR DIFFERENT LOCATION POINTS

Dataset	clem2	clem4	10_w3	mass4	utah2	wisc1	wisc2	wisc3	wisc4
# websites	986	986	986	986	986	986	986	986	986
mean sessions	18	25	357	14	25	18	18	21	25
max sessions	19	25	361	15	25	19	19	22	25
min sessions	18	25	326	14	25	18	18	21	25

D. Preprocessing Data

1) *Flow Identification*: To analyze bidirectional communication, packets are grouped into flows. A flow is defined as a set of packets sharing the same source and destination IP addresses and ports, regardless of direction, following the tuple: $\langle IP_{src}, port_{src}, IP_{dst}, port_{dst} \rangle$.

The code generates a unique and order-independent flow identifier by combining source and destination IP and port pairs, adjusting for directionality. This approach enables the aggregation of all packets within the same bidirectional flow, regardless of the initiating endpoint. In addition, a cumulative relative time column is computed to reconstruct the temporal evolution of each trace, as the original dataset only provides inter-arrival times.

2) *Feature Extraction*: For each flow, a set of statistical features is extracted to characterize the flow's behavior:

- **Packet Counts**: Total number of packets, as well as the number of sent and received packets.
- **Protocol Stats**: Number and ratio of packets using the DoQ protocol.
- **Byte Counts**: Total, sent, and received bytes.
- **Packet Length**: Mean, maximum, minimum, and std of packet lengths for all, sent, and received packets.
- **Temporal Features**: Flow duration and stats of inter-packet times (overall, sent, and received).
- **Throughput**: Packets and bytes per second (overall, sent, and received).
- **Directional Ratios**: Received-sent packet and byte ratios.

These features are computed using vectorized operations to increase efficiency, and the extracted features are aggregated into a structured format.

3) *Batch Processing and Aggregation*: The code iterates through all datasets and websites, extracting features from each CSV trace and aggregating them per website into a single file. This enables scalable processing of large datasets while preserving flow-level statistics for subsequent analysis.

4) *Summary Statistics*: In addition to per-flow feature extraction, we compute summary statistics for each website and capture. This includes the total number of flows and the aggregate duration per website and per capture, facilitating high-level dataset characterization.

This methodology ensures robust and reproducible extraction of flow-level features from raw network traffic, supporting subsequent tasks such as website fingerprinting and traffic analysis. The feature extraction step is central, as it transforms raw packet data into meaningful statistical representations that capture both the temporal dynamics and volumetric properties of network flows.

E. Problem Definition

In the context of encrypted web protocols, we adopt the perspective of a passive adversary aiming to infer the identity of websites visited by a user whose traffic utilizes DoQ.

V. DOQ TRAFFIC ANALYSIS

Despite the encryption, network traffic can exhibit distinctive patterns that can inadvertently reveal information about the underlying user activity. We focus on several features extracted from the flows to analyze the pattern of the traffic.

To illustrate the behavior of the traffic in terms of flow frequency and distribution, we examine how many flows are initiated for each domain across different capture sessions, shown in Fig. 3. Color intensity reflects flow frequency per capture, where consistent activity appears as vertical bands, while irregular patterns indicate sporadic flows. These distributions provide preliminary insights into flow-level features exploitable by fingerprinting models.



Fig. 3. Heatmap showing the number of flows per capture (y-axis) for each domain (x-axis).

A. DoQ Traffic Pattern

In this section, we analyze and compare the patterns of DoQ and non-DoQ traffic to provide a clearer understanding of the distinguishing characteristics of DoQ flows.

Fig. 4 illustrates the evolution of the number of flows across the top 50 websites (ordered by Tranco rank on the x-axis). Beyond flow counts, we examine the nature of these flows with respect to DoQ: (i) flows where DoQ packets comprise 100% of total packets, and (ii) flows with the UDP/853 as

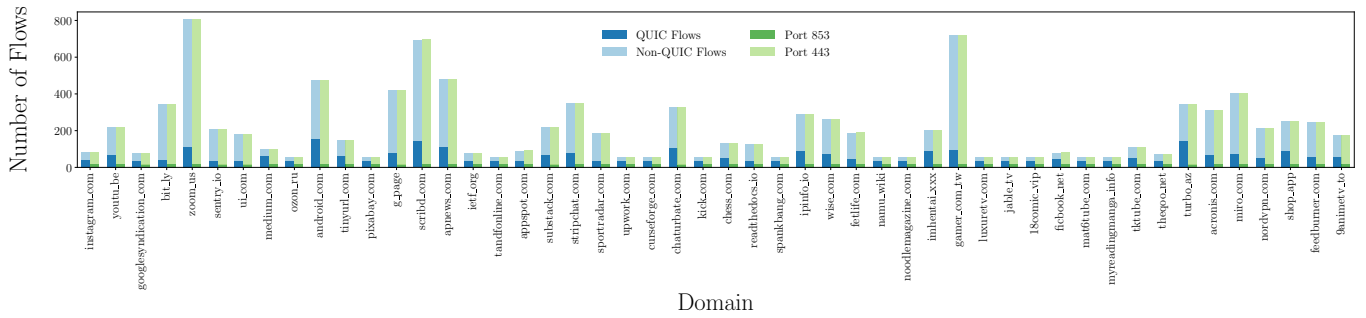


Fig. 4. Stacked bar chart showing the number of flows per destination port for each domain.

the destination port. While flows to port 853 are consistently observed, typically between 18 and 19 per capture, the total number of flows per website varies considerably. For instance, high-activity domains like `zoom.us` and `gamer.com` generate up to 800 flows per session, likely reflecting real-time or background traffic, whereas sites like `18comic` produce as few as 36 flows to port 443, indicating minimal interaction. This variability highlights the diverse traffic behaviors across domains.

The Cumulative Distribution Functions (CDFs) of mean packet sizes for both directions, sent and received, are shown in Fig. 5 for DoQ (top) and non-DoQ (bottom). While the overall distributional patterns are similar across both types, notable differences are observed. Specifically, response packets are consistently larger than sent packets, illustrating the typical asymmetry of DNS exchanges. Moreover, packet sizes over DoQ exhibit a narrower range, indicating more constrained or uniform behavior compared to non-DoQ traffic.

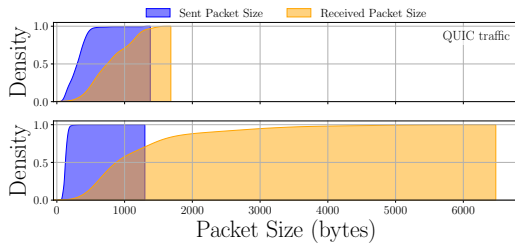


Fig. 5. CDFs of mean packet sizes for queries (sent) and responses (received) for ports 443 (top) and 853 (bottom).

In addition to packet size analysis, we also analyze the duration of flows to assess temporal features. The boxplot of flow durations across multiple domains illustrated in Fig. 6 reveals distinct differences. Across most domains, DoQ flows exhibit longer mean durations. For instance, in `zoom.us`, mean flow duration is approximately 0.75 seconds for non-QUIC traffic compared to 1.76 seconds for DoQ. More pronounced disparities are observed in domains such as `miro.com`, where mean durations are 0.75 seconds versus 2.38 seconds for DoQ and non-DoQ flows, respectively.

DoQ traffic shows longer and more variable flow durations compared to non-DoQ traffic, but with smaller response packet

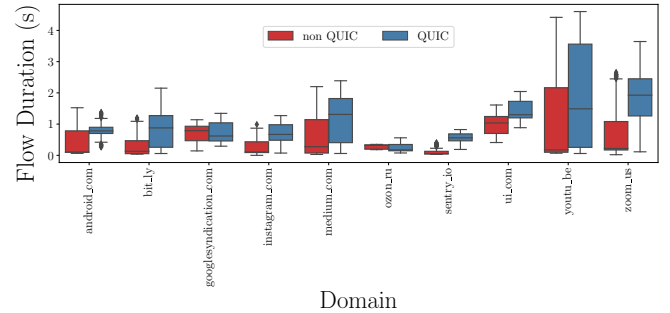


Fig. 6. Boxplot comparing flow durations for QUIC and non QUIC traffic across different domains.

sizes. These contrasting patterns in temporal and size-based features highlight protocol-level differences.

B. DoQ Classification

By examining DoQ traffic properties, we aim to extract insights that reveal how DoQ traffic behaves in practice, and how it can be differentiated from other protocols for the purposes of fingerprinting analysis.

To evaluate which features are most relevant for identifying DoQ traffic, we trained a Random Forest classifier, achieving 100% accuracy on a test set of 6,700 samples. Feature importance analysis revealed that `sent_length_mean` (importance: 0.270) and `sent_bytes` (importance: 0.147) are the most influential features for classification. When retraining the model using only `sent_length_mean`, the classifier still achieved a high accuracy of 91.53%, underscoring its strong discriminative capacity. This result highlights the characteristic size patterns of DoQ traffic. While this section focuses on distinguishing DoQ from other traffic, the next section examines how feature behavior varies across different queried domains.

C. Feature Relevance Analysis

To assess the discriminative influence of each traffic feature, we compute the mutual information (MI) between all the features and the domain label. As shown in Fig. 7, features related to traffic volume and size, such as `received_bytes` (MI = 2.23), `total_bytes` (MI = 2.06), and `recv_length_mean` (MI = 1.57), exhibit the highest MI scores. This suggests that payload volume and distributional statistics of packet

sizes influence the domain level. In contrast, temporal features like duration and time between packets statistics exhibit lower MI values, suggesting they are less informative in isolation. Additionally, the `quic_ratio` feature shows minimal MI, confirming that protocol type alone provides little discriminative value.

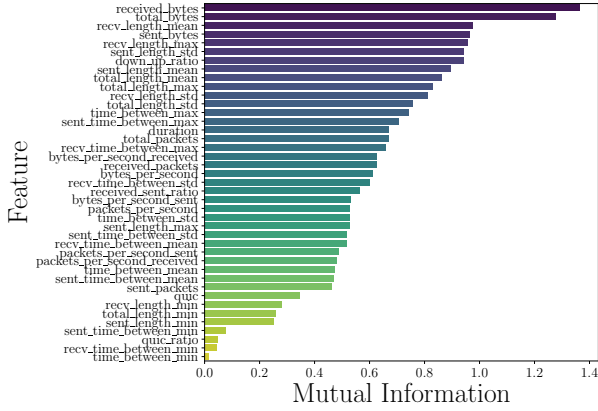


Fig. 7. Mutual Information scores between traffic features and the domain label.

The analysis of DoQ traffic revealed consistent and distinctive characteristics across websites, suggesting their potential utility for fingerprinting. Based on these observations, the next section examines which features contribute most to classification performance and evaluates the effectiveness of machine learning models in WF.

VI. EVALUATION

To assess the feasibility of WF over DoQ traffic, we conduct a comprehensive classification study using various machine learning models. We begin by evaluating performance on a 100-class setting, analyzing accuracy, F1-score, and runtime across models. We then focus on a subset of the top-10 most visited domains to understand behavior on frequently accessed targets. Additionally, we explore dimensionality reduction via Principal Component Analysis (PCA) to examine trade-offs between feature compression and model performance.

A. Performance Results on 100 Websites

Among the evaluated models, XGBoost achieved the highest accuracy of 79.3%, closely followed by Random Forest at 78.5% and Extra Trees at 76.5%. While XGBoost offers the best classification performance, it requires substantially longer training time (1041 seconds) compared to Random Forest (149 seconds) and Extra Trees (45 seconds), indicating a trade-off between accuracy and computational cost. Models like Gradient Boosting and HistGradientBoosting showed notably lower accuracy and, in the case of Gradient Boosting, extremely high training times, making them less practical for this task. KNN, despite its very fast training time, exhibited the poorest accuracy and suffered from very long inference times, limiting its suitability for real-time applications. Overall, tree-based ensemble methods provide the best balance between accuracy and training efficiency.

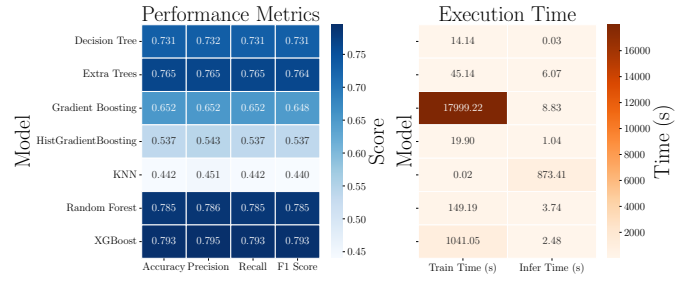


Fig. 8. Performance metrics and computational costs for different models

B. Website Fingerprinting

Zoom into the top-10 most frequently visited websites to understand model behavior on popular targets. We train and evaluate a Random Forest classifier on stratified traffic data using all features, assessing performance via performance metrics and confusion matrix to establish a baseline for encrypted traffic classification.

The confusion matrix presented in Fig. 9 visualizes the performance of the Random Forest classifier in predicting the domain labels for website flows. The results indicate

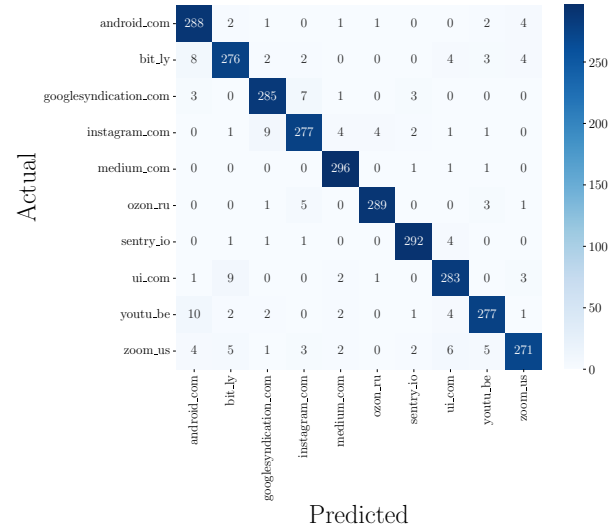


Fig. 9. Confusion matrix of website classification.

strong overall performance, with an average accuracy of 95%. Most classes achieve high precision and recall, particularly for larger classes such as `android_com` and `zoom_us`, which both exceed 90% in F1-score. Some smaller classes, such as `ozon_ru` and `instagram_com`, show lower recall, likely due to limited sample sizes or higher class imbalance. The macro and weighted averages confirm robust classification across classes, with F1-score at 0.95. These results demonstrate the effectiveness of the proposed method in distinguishing between different websites based on network traffic features.

a) *Features Importance:* To analyze the importance of the features, we obtain the features' importance list provided

by the model; this list is illustrated in Fig. 10.

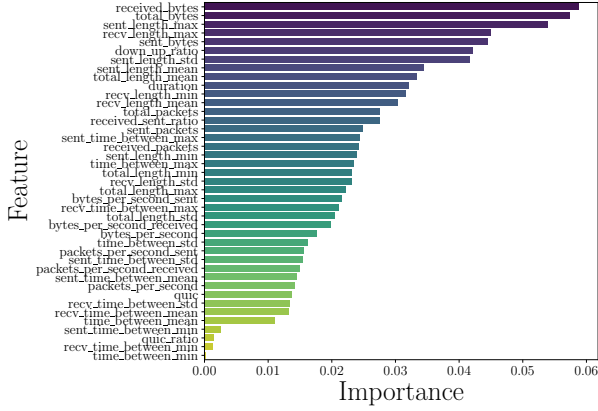


Fig. 10. Features Importance of Random Forest model.

Additionally, we utilize some XAI techniques [20], [21] to interpret the predictions of the model and identify the most influential features. Using a Random Forest, trained on our extracted feature set, SHapley Additive exPlanations (SHAP) provide both global and local explanations by quantifying each feature's contribution to prediction outcomes. The global SHAP summary plot (Fig. 11) highlights that features related to packet size and flow volume have the strongest positive influence on correct domain classification, consistent with the mutual information analysis presented above.

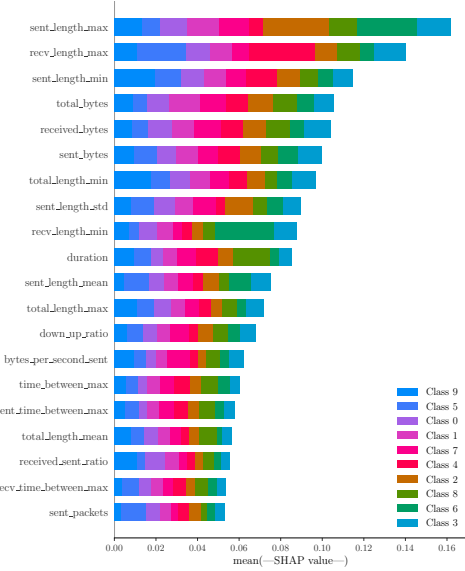


Fig. 11. SHAP values of Random Forest model.

C. PCA-Based Dimensionality Reduction

The objective of this section is to evaluate the trade-off between dimensionality reduction and classification performance using PCA. By projecting the original feature space onto a lower-dimensional subspace, we aim to assess whether similar classification accuracy can be maintained with fewer features,

thereby improving computational efficiency. As illustrated in Fig. 12, the cumulative explained variance increases rapidly with the number of components, reaching over 90% with just 9 principal components.

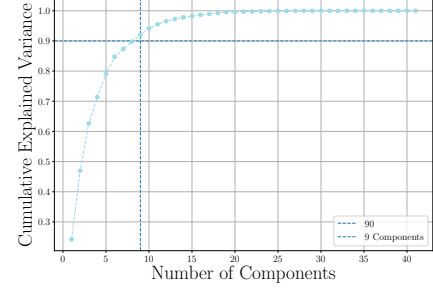


Fig. 12. PCA components.

With PCA reducing features to 9 components, accuracy dropped across models. Extra Trees (0.616) and Random Forest (0.610) remained the top performers but showed lower accuracy than in the full-feature setup. Training times decreased moderately, while simpler models like Decision Tree and k-NN offered faster inference at the cost of performance. Overall, PCA improves efficiency but reduces classification accuracy.

D. Discussion

The classification task becomes notably easier when restricted to the top-10 domains, as reflected in higher accuracy scores and more stable model behavior compared to the 100-domain scenario. Models such as Random Forest and XGBoost show improved precision and reduced confusion among classes due to more consistent and distinguishable traffic patterns. When PCA is applied, a moderate reduction in accuracy is observed, indicating some loss of discriminative information; however, training and inference times improve, particularly for tree-based models. This highlights a trade-off between efficiency and performance, where PCA can reduce energy consumption in resource-constrained settings, it may also compromise accuracy in complex classification tasks.

The results for the 100-class setting case are presented in the Fig.13 (accuracy vs. training time), illustrating the trade-off between performance and efficiency. Reducing training time is essential for enabling the deployment of classification techniques in real-time environments.

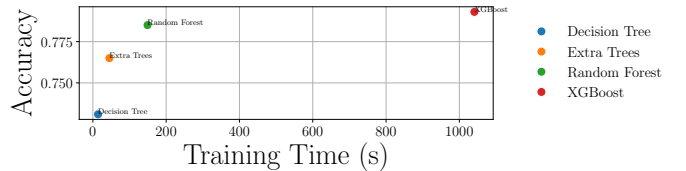


Fig. 13. Accuracy vs training time.

This multi-level evaluation provides a detailed view of classifier effectiveness and the robustness of feature-based identification in encrypted DoQ traffic.

VII. CONCLUSION

DoQ, despite being a promising protocol designed to enhance DNS privacy and performance by encrypting queries and responses over the QUIC transport protocol, remains vulnerable to WF attacks. These vulnerabilities persist primarily due to the inherent leakage of metadata that occurs even under robust encryption. Such metadata leakage enables adversaries to infer valuable information about user traffic, thereby undermining the protocol's fundamental privacy objectives.

This work leverages the dataset introduced in [1], which demonstrated the feasibility of WF attacks on DoQ traffic. In contrast, the present work focuses on understanding the underlying causes of this vulnerability. Our study goes a step further by analyzing how and why these attacks succeed. Specifically, we identify the traffic attributes that contribute most to information leakage. By applying explainable machine learning techniques, we uncover which features are most influential in model predictions, revealing the specific attributes that leak information and are thus most susceptible to exploitation.

The insights gained from this work provide a foundation for designing targeted defenses. Understanding leakage at the feature level can inform mitigation strategies aimed at obfuscating traffic patterns, rather than solely relying on encryption. Such an approach is critical to developing effective countermeasures against information leakage in DoQ traffic. Addressing metadata exposure remains essential for achieving meaningful privacy guarantees in encrypted DNS communication [22].

In summary, despite the significant improvements represented by QUIC in terms of integrated encryption and enhanced performance, important security and privacy concerns remain, particularly fingerprinting vulnerabilities. Our findings emphasize the need for complementary defenses that address metadata leakage. As website fingerprinting remains a realistic and persistent threat, developing practical mitigation techniques for encrypted protocols like DoQ is an essential direction for future work.

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